

Lecture 12: Nonlinear Price Impact, Dynamic Arbitrage, and Monotone Schemes

Lecturer: Andrew Papanicolaou

Scribes: Daniel Chertock

Note: *LaTeX template courtesy of UC Berkeley EECS department.*

Disclaimer: *This note has not been subjected to formal publication review. Distribution outside class requires instructor permission.*

12.1 The No-Dynamic-Arbitrage Principle & Gatheral's Theorem

A foundational requirement for any market microstructure model is the absence of **dynamic arbitrage** (or price manipulation). An execution strategy that starts and ends with zero inventory should not be able to generate positive expected proceeds solely by manipulating the mid-price through its own trading activity.

Definition: Round-Trip Strategy. Let Q_t denote inventory at time $t \in [0, T]$ with continuous trading rate $\nu_t = \frac{d}{dt}Q_t$. An admissible trading strategy $\Pi = \{\nu_t\}_{t \in [0, T]}$ is a **round trip** if initial and terminal inventories are zero,

$$Q_0 = Q_T = 0 \iff \int_0^T \nu_t dt = 0 .$$

Definition: Price Manipulation. Let $C[\Pi]$ denote the expected total cost of executing strategy Π . A round-trip strategy Π is a **price manipulation** (or dynamic arbitrage) if its expected execution cost is strictly negative,

$$\mathbb{E}[C[\Pi]] < 0 .$$

Because $C[\Pi] < 0$ implies positive expected profit from net-zero share accumulation, a market model that permits such strategies is structurally ill-posed.

12.1.1 Impossibility of Pure Nonlinear Permanent Impact

Consider a mid-price S_t subject to exogenous noise and an odd permanent price impact function $g(\nu)$ with $g(-\nu) = -g(\nu)$,

$$dS_t = g(\nu_t) dt + \sigma_S dW_t .$$

Neglecting temporary impact, the expected total cost of a deterministic round trip Π over $[0, T]$ is

$$\mathbb{E}[C[\Pi]] = \mathbb{E} \left[\int_0^T S_t \nu_t dt \right] = \int_0^T \left(S_0 + \int_0^t g(\nu_u) du \right) \nu_t dt .$$

Integrating the S_0 term yields $S_0 \int_0^T \nu_t dt = 0$. Using integration by parts on the double integral,

$$\mathbb{E}[C[\Pi]] = \int_0^T g(\nu_t)(Q_T - Q_t) dt = - \int_0^T g(\nu_t) Q_t dt .$$

Theorem (Gatheral [2]). *Under permanent impact $dS_t = g(\nu_t)dt + \sigma_S dW_t$, the absence of dynamic arbitrage requires $g(\nu)$ to be strictly linear in trading speed ν . Any power-law permanent impact $g(\nu) = \gamma \text{sgn}(\nu)|\nu|^\beta$ with $\beta \neq 1$ allows round-trip arbitrage.*

Proof via Round-Trip Construction. Construct a two-leg round trip over $[0, T]$,

1. **Leg 1 (Buy):** Trade at constant speed $+v_1 > 0$ over $t \in [0, \theta T]$, accumulating inventory $\mathfrak{N} = v_1 \theta T$.
2. **Leg 2 (Sell):** Trade at constant speed $-v_2 < 0$ over $t \in (\theta T, T]$, liquidating inventory $\mathfrak{N} = v_2(1 - \theta)T$.

Matching target volume $\mathfrak{N}$ fixes the time fraction parameter,

$$\theta = \frac{v_2}{v_1 + v_2} \quad \text{and} \quad 1 - \theta = \frac{v_1}{v_1 + v_2} .$$

Under power-law impact $g(\nu) = \gamma \text{sgn}(\nu)|\nu|^\beta$, the expected total round-trip cost evaluates explicitly to

$$\mathbb{E}[C[\Pi]] = \frac{v_1 v_2 T^2}{2(v_1 + v_2)^2} \cdot \gamma \left(v_1 v_2^\beta - v_2 v_1^\beta \right) = \frac{\gamma v_1^2 v_2^2 T^2}{2(v_1 + v_2)^2} \left(v_2^{\beta-1} - v_1^{\beta-1} \right) .$$

We analyze the sign of $\mathbb{E}[C[\Pi]]$ across exponents β ,

- **Sublinear Impact** ($\beta < 1$): The mapping $x \mapsto x^{\beta-1}$ is strictly decreasing. Choosing $v_1 < v_2$ (buying slowly, selling quickly) guarantees $v_2^{\beta-1} - v_1^{\beta-1} < 0$. Thus, $\mathbb{E}[C[\Pi]] < 0$ (positive expected profit).
- **Superlinear Impact** ($\beta > 1$): The mapping $x \mapsto x^{\beta-1}$ is strictly increasing. Choosing $v_1 > v_2$ (buying quickly, selling slowly) guarantees $v_2^{\beta-1} - v_1^{\beta-1} < 0$, yielding $\mathbb{E}[C[\Pi]] < 0$.
- **Linear Impact** ($\beta = 1$): $v_2^0 - v_1^0 = 0$, giving $\mathbb{E}[C[\Pi]] = 0$ for all (v_1, v_2) .

Hence, linear permanent impact ($\beta = 1$) is the *unique* pure permanent impact specification free of price manipulation. ■

12.1.2 Transient Impact Kernels & General No-Arbitrage Conditions

To reconcile empirically observed sublinear impact (e.g., the square-root law $g(\nu) \propto \sqrt{\nu}$) with the absence of dynamic arbitrage, market impact must be modeled as **transient** rather than strictly permanent.

In Gatheral [2], mid-price dynamics are governed by a decay kernel $\Phi(t - s)$,

$$S_t = S_0 + \int_0^t g(\nu_s) \Phi(t - s) ds + \sigma_S W_t .$$

The total expected cost for strategy Π becomes

$$\mathbb{E}[C[\Pi]] = \int_0^T \nu_t \left(\int_0^t g(\nu_s) \Phi(t - s) ds \right) dt .$$

No-Dynamic-Arbitrage Theorem (Gatheral [2]). *The expected execution cost satisfies $\mathbb{E}[C[\Pi]] \geq 0$ for all round trips Π if and only if the kernel $\Phi(\tau)$ is positive semi-definite. For exponential decay $\Phi(\tau) = e^{-\rho\tau}$, no dynamic arbitrage holds if and only if $g(\nu)$ is strictly linear. For power-law decay $\Phi(\tau) = \tau^{-\gamma}$, nonlinear impact $g(\nu) = k \text{sgn}(\nu)|\nu|^p$ is free of dynamic arbitrage if and only if the kernel exponent matches the impact exponent via $\gamma = 1 - p$.*

This fundamental result dictates that if temporary/transient impact is nonlinear in trading rate, its memory must decay via a power law to prevent round-trip arbitrage.

12.2 State-Dependent Price Impact Framework

We now formulate a continuous-time optimal execution framework where price impact depends not only on instantaneous execution volume ν_t , but also on a scalar market state variable $Y_t \in \mathbb{R}$ (e.g., transient impact level, order book depth, or order flow imbalance).

12.2.1 State Dynamics & Cost Functional

Let Q_t denote inventory with liquidation rate $\nu_t = -\frac{d}{dt}Q_t \geq 0$. The mid-price S_t and scalar market state process Y_t evolve according to

$$\begin{aligned} dS_t &= -g(\nu_t, Y_t) dt + \sigma_S dW_t , \\ dY_t &= \mu_Y(Y_t, \nu_t) dt + \sigma_Y dB_t , \end{aligned}$$

where $g(\nu, y)$ represents price impact, $\sigma_S > 0$ and $\sigma_Y > 0$ are constant volatility parameters, and W_t, B_t are standard Brownian motions with correlation $d\langle W, B \rangle_t = \rho dt$.

Continuous execution incurs state-dependent temporary impact $f(\nu_t, Y_t) \geq 0$, giving an effective execution price $\hat{S}_t = S_t - f(\nu_t, Y_t)$. An agent liquidating $Q_0 = \mathfrak{N}$ over $[0, T]$ subject to terminal penalty $G(Q_T, Y_T)$ seeks to maximize expected net proceeds,

$$J(\nu) = \mathbb{E} \left[\int_0^T (S_t - f(\nu_t, Y_t)) \nu_t dt + Q_T S_T - G(Q_T, Y_T) \right] .$$

12.2.2 Derivation of the Reduced HJB Equation

Define the value function $H(t, s, q, y)$ as

$$H(t, s, q, y) = \sup_{\nu \in \mathcal{A}} \mathbb{E}_{t,s,q,y} \left[\int_t^T (S_u - f(\nu_u, Y_u)) \nu_u du + Q_T S_T - G(Q_T, Y_T) \right] .$$

Applying Itô's Lemma and the dynamic programming principle yields the Hamilton-Jacobi-Bellman (HJB) equation,

$$\begin{aligned} \frac{\partial H}{\partial t} + \frac{1}{2} \sigma_S^2 \frac{\partial^2 H}{\partial s^2} + \frac{1}{2} \sigma_Y^2 \frac{\partial^2 H}{\partial y^2} + \rho \sigma_S \sigma_Y \frac{\partial^2 H}{\partial s \partial y} \\ + \max_{\nu \geq 0} \left\{ (s - f(\nu, y)) \nu - \nu \frac{\partial H}{\partial q} - g(\nu, y) \frac{\partial H}{\partial s} + \mu_Y(y, \nu) \frac{\partial H}{\partial y} \right\} = 0 , \end{aligned} \quad (12.1)$$

subject to terminal condition $H(T, s, q, y) = qs - G(q, y)$.

Value Function Decomposition. Because current price s enters execution revenues additively, we decompose the value function into linear price exposure and residual value function $h(t, q, y)$,

$$H(t, s, q, y) = sq + h(t, q, y) .$$

Evaluating partial derivatives,

$$\frac{\partial H}{\partial s} = q , \quad \frac{\partial^2 H}{\partial s^2} = 0 , \quad \frac{\partial H}{\partial q} = s + \frac{\partial h}{\partial q} , \quad \frac{\partial H}{\partial y} = \frac{\partial h}{\partial y} , \quad \frac{\partial^2 H}{\partial y^2} = \frac{\partial^2 h}{\partial y^2} , \quad \frac{\partial^2 H}{\partial s \partial y} = 0 .$$

Substituting these derivatives into equation (12.1) cancels price s identically ($sv - sv = 0$) and eliminates the cross-volatility term since $\frac{\partial^2 H}{\partial s \partial y} = 0$. This reduces the HJB equation to a 2D PDE in (t, q, y) ,

$$\frac{\partial h}{\partial t} + \frac{1}{2}\sigma_Y^2 \frac{\partial^2 h}{\partial y^2} + \max_{\nu \geq 0} \left\{ -f(\nu, y)\nu - qg(\nu, y) - \nu \frac{\partial h}{\partial q} + \mu_Y(y, \nu) \frac{\partial h}{\partial y} \right\} = 0 ,$$

with terminal boundary condition $h(T, q, y) = -G(q, y)$.

Optional Reading: Monotone Schemes for Degenerate HJB Equations

When market impact depends on a scalar state variable Y_t (such as transient impact history I_t with $dI_t = (\nu_t - \rho I_t)dt$), the reduced HJB equation takes the form

$$\frac{\partial h}{\partial t} + \frac{1}{2}\sigma_Y^2 \frac{\partial^2 h}{\partial y^2} + \max_{\nu \geq 0} \left\{ -f(\nu, y)\nu - qg(\nu, y) - \nu \frac{\partial h}{\partial q} + \mu_Y(y, \nu) \frac{\partial h}{\partial y} \right\} = 0 , \quad (12.2)$$

subject to terminal condition $h(T, q, y) = -G(q, y)$.

A. The Degeneracy Problem

Equation (12.2) is **degenerate** because it lacks diffusion in inventory q ($\frac{\partial^2 h}{\partial q^2} = 0$). Standard central finite difference methods create spurious numerical oscillations near gradient kinks. To guarantee local uniform convergence to the unique viscosity solution, we construct a monotone finite difference scheme satisfying the Barles-Souganidis theorem [1] using upwind spatial discretization.

B. Spatial Discretization & Markov Chain Approximation

Discretize state space $(q, y) \in [0, \mathfrak{N}] \times [y_{\min}, y_{\max}]$ on a spatial grid with step sizes $\Delta q, \Delta y$ and time step $\Delta t = T/N$. Let $h_{i,j}^n \approx h(t_n, q_i, y_j)$.

Because selling ($\nu \geq 0$) reduces inventory q , the derivative $\frac{\partial h}{\partial q}$ is discretized using backward (upwind) differences,

$$\mathcal{D}_q^- h_{i,j}^{n+1} = \frac{h_{i,j}^{n+1} - h_{i-1,j}^{n+1}}{\Delta q} .$$

For state variable y , directional derivative $\frac{\partial h}{\partial y}$ is discretized based on the sign of drift $\mu_Y(y_j, \nu)$,

$$\mathcal{D}_y^\pm h_{i,j}^{n+1} = \begin{cases} \frac{h_{i,j+1}^{n+1} - h_{i,j}^{n+1}}{\Delta y} & \text{if } \mu_Y(y_j, \nu) \geq 0 , \\ \frac{h_{i,j}^{n+1} - h_{i,j-1}^{n+1}}{\Delta y} & \text{if } \mu_Y(y_j, \nu) < 0 . \end{cases}$$

The second derivative $\frac{\partial^2 h}{\partial y^2}$ is approximated using standard central differences $\mathcal{D}_{yy}^2 h_{i,j}^{n+1} = \frac{h_{i,j+1}^{n+1} - 2h_{i,j}^{n+1} + h_{i,j-1}^{n+1}}{(\Delta y)^2}$.

Rearranging backward in time gives the Markov Bellman update operator $\mathcal{T}$,

$$h_{i,j}^n = \mathcal{T}(h^{n+1})_{i,j} = \max_{\nu \geq 0} \left\{ \sum_{(i',j')} P_\nu((i,j) \rightarrow (i',j')) h_{i',j'}^{n+1} - \Delta t (f(\nu, y_j)\nu + q_i g(\nu, y_j)) \right\} . \quad (12.3)$$

The transition probabilities P_ν are guaranteed non-negative provided time step Δt satisfies the Courant-Friedrichs-Lewy (CFL) stability condition,

$$\Delta t \cdot \left(\frac{\nu_{\max}}{\Delta q} + \frac{|\mu_Y(y, \nu)|_{\max}}{\Delta y} + \frac{\sigma_Y^2}{(\Delta y)^2} \right) \leq 1 . \quad (12.4)$$

C. Monotonicity & Convergence Proof

Under CFL condition (12.4), all transition probabilities P_ν in update operator (12.3) are non-negative and sum to 1.

Proof of Monotonicity. Let u, v be grid functions such that $u_{i,j} \geq v_{i,j}$ for all (i, j) . Because $P_\nu \geq 0$,

$$\begin{aligned} \mathcal{T}(u)_{i,j} &= \max_{\nu \geq 0} \left\{ \sum_{(i',j')} P_\nu((i,j) \rightarrow (i',j')) u_{i',j'} - \Delta t (f(\nu, y_j)\nu + q_i g(\nu, y_j)) \right\} \\ &\geq \max_{\nu \geq 0} \left\{ \sum_{(i',j')} P_\nu((i,j) \rightarrow (i',j')) v_{i',j'} - \Delta t (f(\nu, y_j)\nu + q_i g(\nu, y_j)) \right\} = \mathcal{T}(v)_{i,j} . \end{aligned}$$

By the Barles-Souganidis Theorem [1], monotonicity guarantees that $h_{i,j}^n$ converges locally uniformly to the unique viscosity solution as $\Delta t, \Delta q, \Delta y \rightarrow 0$. ■

D. Algorithmic Control Extraction

At each grid node (n, i, j) , the optimal execution speed $\nu_{n,i,j}^*$ is extracted by solving the scalar maximization,

$$\nu_{n,i,j}^* = \arg \max_{\nu \geq 0} \left\{ -f(\nu, y_j)\nu - q_i g(\nu, y_j) - \nu \mathcal{D}_q^- h_{i,j}^{n+1} + \mu_Y(y_j, \nu) \mathcal{D}_y^\pm h_{i,j}^{n+1} \right\} .$$

When temporary impact is power-law $f(\nu, y) = k(y)\nu^p$, this 1D optimization is solved efficiently using 3–5 steps of Newton-Raphson root finding on the first-order condition, subject to the non-negativity constraint $\nu \geq 0$.

References

- [1] G. Barles and P. E. Souganidis. Convergence of approximation schemes for fully nonlinear second order equations. *Asymptotic Analysis*, 4(3):271–283, 1991.
- [2] J. Gatheral. No-dynamic-arbitrage and market impact. *Quantitative Finance*, 10(7):749–759, 2010.